

Radioactive decay and nuclear radiation

Atomic structure
Paper 1 · Combined and Separate
26 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 11 starred (★) questions first.

	QUESTION	ANSWER	✓
1	State which part of an atom nuclear radiation is emitted from.	The nucleus	☐
2	Explain why unstable nuclei emit radiation.	To become more stable.	☐
3	What is radioactive decay?	The random process in which an unstable nucleus emits radiation.	☐
4★	Explain why radioactive decay is described as random.	It is impossible to predict which nucleus will decay next, or when.	☐
5★	Define the activity of a radioactive source.	The rate at which its unstable nuclei decay.	☐
6★	State the standard unit of activity.	Becquerel (Bq)	☐
7	What is count-rate?	The number of decays recorded each second by a detector.	☐
8★	Explain why the count-rate is always less than the activity.	The detector picks up only some of the radiation emitted.	☐
9	Name the four types of nuclear radiation.	Alpha particles, beta particles, gamma rays and neutrons	☐
10	Name the three types of nuclear radiation that are ionising.	Alpha, beta and gamma	☐
11★	State what an alpha particle is made of.	Two protons and two neutrons (a helium nucleus)	☐
12★	State what a beta particle is.	A high-speed electron from the nucleus	☐
13★	State what a gamma ray is.	Electromagnetic radiation from the nucleus	☐

QUESTION		ANSWER	✓
14	A nucleus contains no electrons. Describe how it can emit a beta particle.	A neutron turns into a proton and an electron; the electron is emitted.	☐
15 ★	List alpha, beta and gamma radiation from most to least ionising.	Alpha, beta, gamma	☐
16 ★	List alpha, beta and gamma radiation from most to least penetrating.	Gamma, beta, alpha	☐
17 ★	State the range of alpha radiation in air.	A few centimetres	☐
18	State the range of beta radiation in air.	About a metre	☐
19	State the range of gamma radiation in air.	A very long distance (it is hardly absorbed by air)	☐
20	Name the type of radiation stopped by a sheet of paper.	Alpha	☐
21	Name the two types of radiation stopped by a few millimetres of aluminium.	Alpha and beta	☐
22 ★	State what is needed to absorb most gamma radiation.	Several centimetres of lead	☐
23	Name the type of radiation stopped by skin.	Alpha	☐
24	Name a detector of nuclear radiation.	A Geiger–Müller tube	☐
25	State the relative charges of alpha, beta and gamma radiation.	Alpha: +2 Beta: –1 Gamma: 0	☐
26	What is ionisation?	An atom gaining or losing electrons to become an ion.	☐